

#This code will be used to demonstrate proposition 7.20. We will demonstrate that $m2$ is increasing along the $\mathcal{V}$ curve where $m2$ is positive with $m1$

#Define M2;

$$M2 := \frac{\frac{y}{8} \cdot (s^{-3} - T) \cdot (z^{-3} - w^{-3})}{\frac{x}{4} \cdot ((w^{-3} - x^{-3}) \cdot (y^{-3} - z^{-3}) + (z^{-3} - x^{-3}) \cdot (y^{-3} - w^{-3}))} :$$

$$s := \frac{y}{2} :$$

$$T := \frac{z^{-3} + w^{-3}}{2} :$$

#Apply the rational parametrization constructed in this chapter 4.

$$\text{parametrization} := \left\{ w = a + \frac{1}{a}, \right.$$

$$z = b + \frac{1}{b},$$

$$x = a - \frac{1}{a} + b - \frac{1}{b},$$

$$y = a - \frac{1}{a} - b + \frac{1}{b} \left. \right\} :$$

$$m2 := \text{simplify}(\text{expand}(\text{subs}(\text{parametrization}, M2))) ;$$

$$\begin{aligned} m2 := & - \left((a - b)^2 (b a - 1)^3 (b a + 1) (a^{12} b^6 + 3 (-b^7 + b^5) a^{11} + 3 (b^8 - 2 b^6 + b^4) a^{10} \right. \\ & + (-16 b^9 - 45 b^7 - 45 b^5 - 14 b^3) a^9 + 3 (-b^{10} - 5 b^6 + 4 b^4 + b^2) a^8 + 3 (b^{11} - 17 b^9 \\ & - 48 b^7 - 58 b^5 - 15 b^3 + b) a^7 + (-b^{12} + 6 b^{10} + 15 b^8 - 15 b^4 - 6 b^2 + 1) a^6 + 3 (\\ & - b^{11} - 17 b^9 - 38 b^7 - 48 b^5 - 15 b^3 - b) a^5 + 3 (-b^{10} - 4 b^8 + 5 b^6 + b^2) a^4 + (-18 b^9 \\ & - 51 b^7 - 51 b^5 - 16 b^3) a^3 + 3 (-b^8 + 2 b^6 - b^4) a^2 + 3 (-b^7 + b^5) a - b^6) (b \\ & + a)^2 (a^4 b^2 + (b^3 + b) a^3 + (b^4 + 4 b^2 + 1) a^2 + (b^3 + b) a + b^2) \Bigg) \Bigg/ \left(48 a^3 (b^2 \right. \\ & + 1)^3 \left((b^8 - b^6 + b^4) a^{10} - \frac{b^2 (b^8 - 2 b^6 + 8 b^4 - 2 b^2 + 1) a^8}{2} \right. \\ & + \frac{b (b^4 - 4 b^2 + 1) (b^2 + 1)^3 a^7}{2} + \frac{(b^4 + 1) (b^8 + 8 b^4 + 1) a^6}{6} \\ & - \frac{b (b - 1)^2 (b + 1)^2 (b^2 + 1)^3 a^5}{2} - \frac{b^2 (b^8 + 2 b^6 - 2 b^4 + 2 b^2 + 1) a^4}{2} \\ & \left. \left. - \frac{b^3 (b^2 + 1)^3 a^3}{3} + a^2 b^6 - \frac{b^6}{3} \right) b^3 (a^2 + 1)^3 \right) \end{aligned} \quad (1)$$

#Define r4

$$\begin{aligned} r4 := & (a^{22} b^{12} + (8 b^{13} - 6 b^{11}) a^{21} + (30 b^{14} + 6 b^{12} + 15 b^{10}) a^{20} + (74 b^{15} + 162 b^{13} - 102 b^{11} \\ & - 18 b^9) a^{19} + (145 b^{16} + 600 b^{14} + 93 b^{12} + 198 b^{10} + 3 b^8) a^{18} + (252 b^{17} + 1380 b^{15} \\ & + 1884 b^{13} - 762 b^{11} - 96 b^9 + 24 b^7) a^{17} + (145 b^{18} + 1092 b^{16} + 1908 b^{14} - 2920 b^{12} \\ & - 3162 b^{10} - 1740 b^8 - 294 b^6) a^{16} + (74 b^{19} + 1380 b^{17} + 4692 b^{15} + 3340 b^{13} - 8010 b^{11} \\ & - 2472 b^9 - 132 b^7 + 24 b^5) a^{15} + (30 b^{20} + 600 b^{18} + 1908 b^{16} - 3516 b^{14} - 24921 b^{12} \\ & - 26352 b^{10} - 11586 b^8 - 1740 b^6 + 3 b^4) a^{14} + (8 b^{21} + 162 b^{19} + 1884 b^{17} + 3340 b^{15} \\ & - 8268 b^{13} - 34416 b^{11} - 14520 b^9 - 2472 b^7 - 96 b^5 - 18 b^3) a^{13} + b^2 (b^{20} + 6 b^{18} + 93 b^{16} \\ & - 2920 b^{14} - 24921 b^{12} - 68232 b^{10} - 67416 b^8 - 26352 b^6 - 3162 b^4 + 198 b^2 + 15) a^{12} \\ & - 6 b (b^{20} + 17 b^{18} + 127 b^{16} + 1335 b^{14} + 5736 b^{12} + 11692 b^{10} + 5736 b^8 + 1335 b^6 + 127 b^4 \\ & + 17 b^2 + 1) a^{11} + (15 b^{20} + 198 b^{18} - 3162 b^{16} - 26352 b^{14} - 67416 b^{12} - 68232 b^{10} \\ & - 24921 b^8 - 2920 b^6 + 93 b^4 + 6 b^2 + 1) a^{10} + (-18 b^{19} - 96 b^{17} - 2472 b^{15} - 14520 b^{13} \\ & - 34416 b^{11} - 8268 b^9 + 3340 b^7 + 1884 b^5 + 162 b^3 + 8 b) a^9 + 3 b^2 (b^{16} - 580 b^{14} \\ & - 3862 b^{12} - 8784 b^{10} - 8307 b^8 - 1172 b^6 + 636 b^4 + 200 b^2 + 10) a^8 + (24 b^{17} - 132 b^{15} \\ & - 2472 b^{13} - 8010 b^{11} + 3340 b^9 + 4692 b^7 + 1380 b^5 + 74 b^3) a^7 + (-294 b^{16} - 1740 b^{14} \\ & - 3162 b^{12} - 2920 b^{10} + 1908 b^8 + 1092 b^6 + 145 b^4) a^6 + (24 b^{15} - 96 b^{13} - 762 b^{11} \\ & + 1884 b^9 + 1380 b^7 + 252 b^5) a^5 + (3 b^{14} + 198 b^{12} + 93 b^{10} + 600 b^8 + 145 b^6) a^4 + (-18 b^{13} - 102 b^{11} + 162 b^9 + 74 b^7) a^3 + (15 b^{12} + 6 b^{10} + 30 b^8) a^2 + (-6 b^{11} + 8 b^9) a \\ & + b^{10}) : \end{aligned}$$

#Using the implicit function theorem **and** the chain rule, the derivative of m2 is zero
if the function f constructed below is zero.

$$f := -\text{diff}(r4, b) \cdot \text{diff}(m2, a) + \text{diff}(m2, b) \cdot \text{diff}(r4, a) :$$

#Calculate the numerator of f

$$\tilde{f} := \text{numer}(f) :$$

#Plot the curves r4=0 and $\tilde{f}=0$.

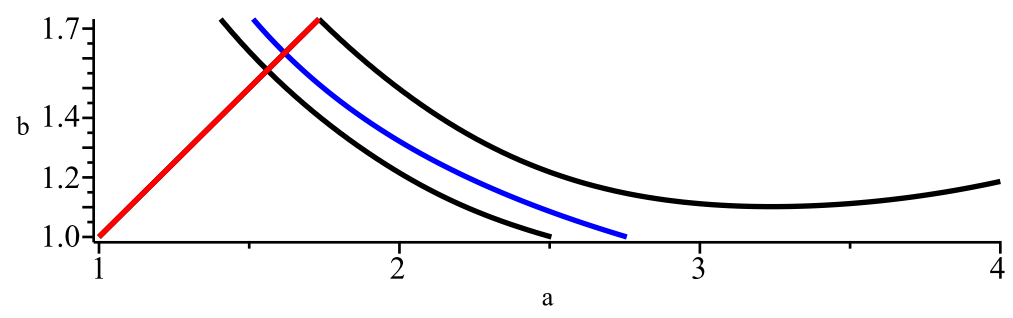
$$r := a - b :$$

$$v := \frac{1 + \text{sqrt}(5)}{2} :$$

with(plots) :

implicitplot([r4=0, $\tilde{f}=0$, r=0], a = 1 .. 4, b = 1 .. (sqrt(3)),
scaling = constrained,
gridrefine = 2,
thickness = 2,
color = [blue, black, red],
labels = ["a", "b"],

title = "Graphic for r4=0 and $\tilde{f}=0$ ";
Graphic for r4=0 and $\tilde{f}=0$



#Let's build the DescartesSingChanges function that counts the number of sign changes between the coefficients of a given polynomial.

```
DescartesSignChanges := proc (polynomial, variable)  
  local coeffs, nonzero_coeffs, s, i, n;  
  
  # Get the list of coefficients from highest degree to constant term  
  coeffs := [seq(coeff (polynomial, variable, i), i = degree (polynomial, variable) .. 0, -1)];  
  
  # Remove zero coefficients (they do not count in Descartes' Rule)  
  nonzero_coeffs := remove (c → c = 0, coeffs);  
  
  # Count the number of sign changes  
  n := nops (nonzero_coeffs);  
  s := 0;  
  for i from 1 to n - 1 do  
    if signum (nonzero_coeffs[i]) * signum (nonzero_coeffs[i + 1]) = -1 then  
      s := s + 1;  
    end if;  
  end do;  
  
  return s;  
end proc;
```

#Compute the resultant between $r4$ and $\tilde{f}$ with respect to a

$R := \text{resultant}(r4, \tilde{f}, a) :$

#Compute the factors of the resultant R and count how many there are

$fator := [op(factor(R))]:$
 $nops(fator);$

9

(2)

#We list below the factors of the resultant R , except for the factor 8 because it is too large. When necessary, we will use Sturm's algorithm to check if the factors cancel each other out within the interval $[1, v)$ where γ is defined. For the $r8$ factor, because of its high degree, we will use Moebius transformations as it requires less processing power.

$r1 := fator[1];$

$r1 :=$

(3)

$-237213776013387785127790706222196476499439870168868563287857307928933368203 \backslash$
 $088194342587495659654447729958868484096$

$r2 := fator[2];$

$r2 := b^{660}$

(4)

$r3 := fator[3];$

$r3 := b^2 + b - 1$

(5)

$r4 := fator[4];$

$r4 := b^2 - b - 1$

(6)

$r5 := fator[5];$

$r5 := (b^2 + 1)^{444}$

(7)

$r6 := fator[6];$

$r6 := (b^4 + 2b^3 + 6b^2 + 2b + 1)^2$

(8)

$r7 := fator[7];$

$\text{sturm}(\text{sturmseq}(r7, b), b, 1, (v));$ #number of roots in $[1, v]$

$r7 := (b^4 - 2b^3 + 6b^2 - 2b + 1)^2$

0

(9)

$r9 := fator[9];$

$r9 := (b^4 + 34b^2 + 1)^{12}$

(10)

#The factor $r8$ has degree 316, so we will not show it, just compute how many zeros it has in the desired range.

$r8 := fator[8];$

```
degree( r8, b);
```

316

(11)

```
#Apply the Möbius transformation on the interval  $\left(1, \frac{1 + \sqrt{5}}{2}\right)$ , constructing R8
```

```
moebius :=  $\left\{ b = \frac{1 + v \cdot d}{1 + d} \right\} :$ 
```

```
R8 := collect(numer(expand(subs(moebius, r8))), d) :
```

```
#From here we will analyze the signal variation of all coefficients R8
```

```
DescartesSignChanges(R8, d);
```

0

(12)